

DECREASE AND CONQUER

Decrease by a Constant ($n - k$)



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INFORMATION TECHNOLOGY

KEY IDEA

- **Instead of reducing the problem by 1 element at a time ($n - 1$), we reduce it by a fixed constant k .**
- **At each step:**
 - Process or eliminate k elements
 - Reduce problem size from n to $n - k$
 - Solve the same problem on the smaller input
- **Only one smaller subproblem remains each time.**





MAXIMUM OF AN ARRAY (PROCESS 3 ELEMENTS AT A TIME)

- **Problem**

- Find the maximum value in the array:
 - $A = [17, 4, 29, 8, 21, 35, 6, 14]$
 - Let
 - $k = 3$

- **Idea ($n - k$)**

- Examine k elements at a time
- Keep only the best candidate
- Reduce problem size by k
- Repeat on the remaining elements





MAXIMUM OF AN ARRAY (PROCESS 3 ELEMENTS AT A TIME) (CONT.)

- **Step 1 – First k elements**
 - $[17, 4, 29] \rightarrow \text{max} = 29$
- **Candidate max = 29**
- **Remaining elements:**
 - $[8, 21, 35, 6, 14]$
- **Problem size: 8 $\rightarrow$ 5**





MAXIMUM OF AN ARRAY (PROCESS 3 ELEMENTS AT A TIME) (CONT.)

- **Step 2 – First k elements**
 - $[8, 21, 35] \rightarrow \text{max} = 35$
- **Compare with previous max (29) $\rightarrow$ update max = 35**
- **Remaining elements:**
 - $[6, 14]$
- **Problem size: 5 $\rightarrow$ 2**





MAXIMUM OF AN ARRAY (PROCESS 3 ELEMENTS AT A TIME) (CONT.)

- **Step 3 – First k elements**
 - $[6, 14] \rightarrow \text{max} = 14$
- **Compare with previous max (35) $\rightarrow$ keep max = 35**
- **FINAL ANSWER**
 - **MAXIMUM = 35**



WHY $N - K$?

- Each step reduces the problem by 3 elements
- Only one smaller subproblem remains
- Same logic applied repeatedly





COUNTING A TARGET VALUE (SKIP BY K)

- **Problem**

- Count how many times the value 10 appears in:
 - $A = [10, 3, 10, 4, 10, 8, 7, 10]$
 - Let
 - $k = 2$

- **Idea**

- Check $k = 2$ elements at a time
- Count occurrences
- Reduce the remaining list by k





COUNTING A TARGET VALUE (SKIP BY K) (CONT.)

- **Step 1**
 - $[10, 3] \rightarrow \text{count} = 1$
 - Remaining: $[10, 4, 10, 8, 7, 10]$





COUNTING A TARGET VALUE (SKIP BY K) (CONT.)

- **Step 2**
 - $[10, 4] \rightarrow \text{count} = 2$
 - Remaining: $[10, 8, 7, 10]$





COUNTING A TARGET VALUE (SKIP BY K) (CONT.)

- **Step 3**
 - $[10, 8] \rightarrow \text{count} = 3$
 - Remaining: $[7, 10]$





COUNTING A TARGET VALUE (SKIP BY K) (CONT.)

- **Step 4**
 - $[7, 10] \rightarrow \text{count} = 4$





COUNTING A TARGET VALUE (SKIP BY K) (CONT.)

- **FINAL ANSWER**
 - Number of occurrences of 10 = 4





DECREASE-AND-CONQUER OBSERVATION

- Each iteration reduces the problem size from $n \rightarrow n - 2$.





PARTIAL SUM OF AN ARRAY (BATCH PROCESSING)

- **Problem**
 - Compute the sum of the array:
 - $A = [5, 2, 9, 4, 7, 1, 6, 3]$
 - Let
 - $k = 4$





PARTIAL SUM OF AN ARRAY (BATCH PROCESSING) (CONT.)

- **Step 2 – First k elements**
 - $[7, 1, 6, 3] \rightarrow \text{sum} = 17$





PARTIAL SUM OF AN ARRAY (BATCH PROCESSING) (CONT.)

- **FINAL ANSWER**
 - Total = $20 + 17 = 37$



WHY $N - K$?

- Each recursive/iterative step removes 4 elements
- Same operation repeated on smaller input





SEARCHING WITH K-STEP ELIMINATION (LINEAR SEARCH VARIANT)

- **Problem**
 - Search for key = 18 in:
 - $A = [9, 14, 3, 21, 18, 27, 6]$
 - Let
 - $k = 2$





SEARCHING WITH K-STEP ELIMINATION (LINEAR SEARCH VARIANT) (CONT.)

- **STEP 1**
 - $[9, 14] \rightarrow$ not found
- **Remaining: $[3, 21, 18, 27, 6]$**





SEARCHING WITH K-STEP ELIMINATION (LINEAR SEARCH VARIANT) (CONT.)

- **STEP 2**
 - $[3, 21] \rightarrow$ not found
- **Remaining: $[18, 27, 6]$**





SEARCHING WITH K-STEP ELIMINATION (LINEAR SEARCH VARIANT) (CONT.)

- **STEP 3,**
 - $[18, 27] \rightarrow \text{FOUND}$
- **FINAL ANSWER**
 - KEY FOUND





DECREASE-AND-CONQUER EXPLANATION

- **This is a generalized linear search where:**
 - Problem reduces by $k = 2$ each step
 - Search logic remains the same





ARRAY EQUALITY CHECK (BLOCK COMPARISON)

- **Problem**
 - Check if all values in the array are equal:
 - $A = [7, 7, 7, 7, 7, 7]$
 - Let
 - $k = 3$





ARRAY EQUALITY CHECK (BLOCK COMPARISON) (CONT.)

- **Step 1**
 - $[7, 7, 7] \rightarrow$ all equal
- **Remaining: $[7, 7, 7]$**





ARRAY EQUALITY CHECK (BLOCK COMPARISON) (CONT.)

- **Step 2**
 - $[7, 7, 7] \rightarrow$ all equal
- **Results**
 - All values are equal



END



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